

LINGZHI YUAN

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🎓 EDUCATION

University of Maryland, College Park, MD, USA Aug. 2025 – Present

Ph.D in Computer Science, expected May 2030

Zhejiang University, Hangzhou, China

Sep. 2021 – Jun. 2025

B.Eng in Automation, *GPA*: 3.96/4, *Rank*: 1/58

Minor in Intensive Training Honors Program (ITP) at [Chu Kochen Honors College](#) (Top 40/5400)

💼 EXPERIENCE

Furong Huang Lab, University of Maryland, College Park, MD, USA Aug. 2025 – Present

Research Assistant, focused on Trustworthy ML and LLM/Agent Safety advised by Prof. [Furong Huang](#)

UNIUBI AI Hangzhou, China

Mar. 2025 – Jun. 2025

Student Intern, focused on VLM's Instruction Following and Data Augmentation

Secure Learning Lab, The University of Chicago, Chicago, IL, USA

Mar. 2024 – Dec. 2024

Research Assistant, focused on Trustworthy ML and Multi-modal Safety under the guidance of Prof. [Bo Li](#)

Nanyang Technological University, Singapore

Aug. 2023 – Sep. 2023

Summer School Participant, Machine Learning and Its Applications program led by Prof. [Kezhi Mao](#)

📄 PUBLICATIONS

* Denotes equal contribution

Meta²: Precomputing Reusable Knowledge for Online Harness Evolution in Long-Horizon Agents

Lingzhi Yuan*, Yiguang Cai*, Weize Liu, Feijiang Han, Yongkyun Lee, Udari Madhushani Sehwag, Furong Huang. *Under Review*

Hypothesis-Conditioned Evaluation for Efficient Harness Optimization

Lingzhi Yuan*, Zhi Zhang*, Arash A. Amini, Furong Huang. *Under Review*

LatentShield: Composable Latent Multi-Agent Systems with Decision-Boundary Monitoring

Baicheng Chen*, Anirudh Satheesh*, [Lingzhi Yuan](#), Drew Gjerstad, Andrew Mendz, Ami Britti Antoni Gnanaaraj, Mingyu Kang, Benjamin Leinwand, Andrew Adams Schoen, Furong Huang. *Under Review*

FlowBank: Query-Adaptive Agentic Workflow Optimization through Precompute-and-Reuse

[Lingzhi Yuan](#), Chenghao Deng, Fangxu Yu, Souradip Chakraborty, Mohammad Rostami, Furong Huang. *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*

PropmtGuard: Soft Prompt-Guided Unsafe Content Moderation for Text-to-Image Models

Lingzhi Yuan*, Xinfeng Li*, Chejian Xu, Guan hong Tao, Xiaojun Jia, Yihao Huang, Wei Dong, Yang Liu, Bo Li. *The IEEE Transactions on Information Forensics and Security (TIFS)*

TSRBench: A Comprehensive Multi-task Multi-modal Time Series Reasoning Benchmark for Generalist Models

Fangxu Yu, Xingang Guo, [Lingzhi Yuan](#), Haoqiang Kang, Hongyu Zhao, Lianhui Qin, Furong Huang, Bin Hu, Tianyi Zhou. *Forty-Third International Conference on Machine Learning (ICML), 2026*

MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models

Chejian Xu*, Jiawei Zhang*, Zhaorun Chen*, Chulin Xie*, Mintong Kang*, Zhuowen Yuan*, Zidi Xiong*, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, Peiyang Xu, Chengquan Guo, Andy Zhou, Jeffrey Ziwei Tan, Zhun Wang, Alexander Xiong, Xuandong Zhao, Yu Gai, Francesco Pinto, Yujin Potter, Zhen Xiang, Zinan Lin, Dan Hendrycks, Dawn Song, Bo Li. *International Conference on Learning Representation (ICLR)*, 2025

📁 SELECTED PROJECTS

Query-Adaptive Agentic Workflow Optimization

Sep. 2025 - May. 2026

Project Leader Advisor: Prof. [Furong Huang](#), Associate Professor at University of Maryland

Developed a portfolio-based framework that precomputes a bank of complementary agentic workflows offline and adaptively routes each query to the best one at inference time.

- Revealed workflow complementarity via motivating analysis: discarded task-level workflows retain reuse value, and query-level generation gains can be recovered by cheaper precomputed workflows.
- Designed a three-stage framework combining complementarity-oriented MCTS search, coverage-aware portfolio curation, and GNN-based query-workflow matching.
- Achieved the best average score across five benchmarks, improving over the strongest automated and hand-crafted baselines by 4.26% and 14.92% relative.

Soft Prompt-Guided Defense for Text-to-Image Models

Jun. 2024 - Dec. 2024

Project Leader Advisor: Prof. [Bo Li](#), Associate Professor at University of Chicago/UIUC

Developing an effective and efficient framework to mitigate NSFW content generation across different unsafe categories by T2I models.

- Conducted an extensive survey of advanced safety protection techniques for T2I models, identifying critical challenges and areas for improvement.
- Designed and implemented an advanced framework addressing multiple categories of unsafe content generation, achieving state-of-the-art performance in NSFW moderation with high time efficiency.
- Authored and submitted a manuscript for peer review.

Comprehensive Assessment of Trustworthiness in Multimodal Models

Mar. 2024 – Oct. 2024

Research Assistant Advisor: Prof. [Bo Li](#), Associate Professor at University of Chicago/UIUC

Conduct a comprehensive evaluation of the safety and trustworthiness of multimodal foundation models, addressing key challenges in robustness and reliability.

- Investigated adversarial robustness by applying cutting-edge red-team algorithms, such as MMP and GCG for Text-to-Image (T2I) models and Attack-VLM for Image-to-Text (I2T) models, on surrogate architectures to develop a robust benchmark dataset.
- Evaluated state-of-the-art multimodal models including Dalle-3, GPT-4o etc. against the curated dataset, providing a detailed analysis of their vulnerability and resilience under adversarial conditions.
- Authored and submitted a manuscript on the findings to **ICLR 2025** for peer review.

⚙️ SKILLS

- **Programming Languages:** solid expertise in Python, Matlab, C++, and various algorithms
- **Tools:** PyTorch, Git, Linux/Unix, Transformers, Diffusers, OpenCV
- **Language Proficiency:** TOEFL 105 (Reading: 30 Listening: 27 Speaking: 23 Writing: 25)

🏆 HONORS AND AWARDS

University of Maryland Graduate School Dean's Fellowship

Aug. 2025

Infineon Power Semiconductor Scholarship

Sep. 2024, Sep. 2023

Zhejiang University Second Class Scholarship

Sep. 2022, Sep. 2023

Nitori International Scholarship

Sep. 2022